Learning to Manipulate a New Object from Action-Free Videos

Group 9
McGill University
ECSE 552 — Project proposal

Background

Application. We teach a robot arm that manipulates one object to handle new ones from videos alone: warehouse and home robots meet new items constantly, and recording paired images and motor commands for each requires operating the robot. We study this in OCBench simulation (UR5e arm, Robotiq gripper) [1].

Problem and need. Visuomotor policies imitate images paired with motor commands, and such pairs are costly: 76k teleoperated demonstrations took 50 people a year [2]. Videos without commands are cheap (e.g., generated by world models or video generation models [3]). An inverse dynamics model (IDM), trained on few labeled demonstrations [4], infers the command between consecutive frames and labels such videos for imitation [5, 6]. Here the target videos show new objects, unlike the IDM's training data. Formally, given labeled cube transitions $\mathcal{D}_A = \{(o_t, o_{t+1}, a_t)\}$ and unlabeled target videos $\mathcal{D}_B = \{o_t\}$, we seek an image-only policy $\pi(a \mid o)$ that succeeds on the new objects under R1–R6.

Constraints. Red values are still to be agreed.

ID	Requirement	Definition	Target
R1	Supervision	Target data are RGB frames only	Zero target labels
R2	Label budget	Labeled cube demonstrations, identical for every method	300 demos
R3	Inputs	No simulator state in any learned model	Images only
R4	Target	Success gain over the cube-only policy on each target, as a fraction of the oracle's gain	$\geq$ 50% of gap
R5	Retention	Drop in cube success after adding target data	$\leq$ 10 points
R6	Real time	Policy inference per call, within one 50 Hz control step (IDM offline)	$<$ 20 ms

Aims/Goals

Goals. Meet R4 and R5 on each new object with an image-only policy trained on cube demonstrations and action-free target videos; R1–R3 hold by design and R6 is measured.

Scope. We use OCBench's visual single-block task; the source is its default $6 \times 6 \times 6$ cm red cube. The targets separate the factors: (A) a $6 \times 6 \times 6$ cm blue cube (appearance only), (B) a $9 \times 6 \times 6$ cm red block (geometry, same grasp width), (C) a $9 \times 7 \times 6$ cm red block (geometry and grasp width), and, optionally, (D) a $9 \times 7 \times 6$ cm blue block (all combined). Expert grasps are adapted to each; robot, cameras, lighting, and task stay fixed. Human videos, real hardware, and large pretrained models are out of scope.

Proposed solution. We adapt the IDM to each target before labeling its videos. The IDM sees frame differences, which emphasize the unchanged arm, and domain-adversarial training [7] makes its features indistinguishable between cube and target frames while still predicting cube actions. One policy per target then trains on cube demonstrations plus pseudo-labeled target videos. Adaptation should help most on A, whose actions match the cube's.

Methodology

Design and implementation. Observations are 224×224 images from the front, side, and wrist cameras; actions are seven normalized joint and gripper increments at 50 Hz. The IDM's convolutional encoder, shared across cameras, embeds o_t, o_{t+1}, and their difference, and an MLP regresses a_t with mean squared error. A discriminator classifies these features as cube or target (binary cross-entropy); a gradient reversal layer returns its gradient to the encoder with flipped sign, scaled by a weight λ that ramps up as in DANN; $\lambda = 0$ gives the unadapted IDM. The policy shares the encoder, takes o_t, and is trained with flow matching to predict the next 25 actions, executed open loop as in OCBench [1]. All methods share encoder, data, and budget (PyTorch).

Validation, testing, and evaluation. Splits are made by episode. Target actions never select models: the λ schedule is fixed in advance, IDM checkpoints follow source validation error, and policies train for a fixed number of steps. We report target success, cube retention, per-component IDM error, and latency over three seeds and 100 episodes per method and object, with identical initial states.

Alternatives. We compare (a) a cube-only behavioral cloning policy, (b) an unadapted IDM, (c) our adversarial IDM, (d) an oracle trained with true target actions, and (f) the cube-only policy trained with strong augmentation (full-range hue, brightness, contrast, and saturation jitter, and small random crops) and no target videos; optionally, (e) applies the alignment to the policy encoder instead. (b) versus (c) isolates adaptation; (f) versus (c) tests whether target videos are needed at all; (a) and (d) anchor R4. We expect (f) to be competitive on A but not on B and C: augmentation only covers changes we can anticipate and synthesize, such as color, whereas target videos show the actual new object, including geometry changes that no image transformation produces. (c) should also outperform (b), since an unadapted IDM has never seen the target object and may mislabel its frames.

Feasibility and resources

Dataset, resources, and plan. We generate all data in OCBench with MuJoCo: 300 labeled cube demonstrations and 200 action-free videos per target, keeping true target actions only for evaluation and the oracle; no licensing or ethics constraints apply. Training uses one NVIDIA A10G GPU; the pilot estimates compute. A pilot by October 21 checks expert success and, on the cube and each target, the cube-only policy's success and the IDM's error. Core methods follow by November 1; evaluation by November 18.

Risk analysis

Pitfalls and mitigation. Adversarial training aligns feature distributions, not the action each feature maps to [8]; we expect actions to stay readable from arm motion, which per-component errors will test. If the expert fails on a target, we adjust the object or grasp; if the cube-only policy fails on the cube, we use OCBench's shorter lite variant. If alignment hurts, we report negative transfer; optional experiments (D, e) go first if time runs short.

Role of group members

All members collaborate and review the report.

Member	Role	Tasks
MEMBER 1	Data and evaluation	Simulation, expert demos, datasets, splits, rollouts, metrics
MEMBER 2	Adversarial adaptation	Discriminator, adversarial training, tuning, experiments
MEMBER 3	Inverse dynamics model	IDM architecture and training, pseudo-labeling, error analysis
MEMBER 4	Baselines and alternatives	Policy training, baselines, alternative methods, comparisons

Generative-AI disclosure. Codex (OpenAI) and Claude Code (Anthropic) assisted with literature search, bibliography formatting, document structure, and prose. We used them to survey recent work and to format the document; the team revised all text and verified every claim and citation against the original sources.

References

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